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Effects of tomato treatments with commercial formulations of entomopathogenic fungi on the pest *Tuta absoluta* and the predator *Macrolophus pygmaeus*

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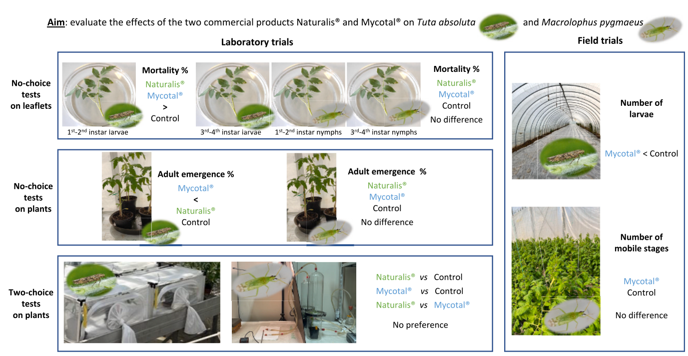
HIGHLIGHTS

- Effect of Naturalis® and Mycotal® was evaluated on pest and predator biology and behaviour.
- Naturalis® and Mycotal® increased mortality of only younger *T. absoluta* larvae.
- Mycotal® reduced development rate of *T. absoluta.*
- Pest and predator showed no preference between fungus-treated or untreated plants.
- In field trial, Mycotal® reduced significantly *T. absoluta* population.

GRAPHICAL ABSTRACT



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ABSTRACT

Tuta absoluta is a devasting pest causing severe yield losses in most tomato-growing areas. *Macrolophus pygmaeus* is a generalist predator released in tomato crop to control *T. absoluta*. Entomopathogenic fungi are also used on tomato against other pests, but they can play a role in controlling *T. absoluta*. Therefore, the effects of two commercial products Naturalis® and Mycotal®, based on the entomopathogenic fungi *Beauveria bassiana* and *Akanthomyces muscarius*, respectively, on *T. absoluta* and *M. pygmaeus* were assessed. In no-choice tests on leaflets, mortality was significantly higher on fungus-treated leaflets for 1^{st}–2^{nd} instar *T. absoluta* larvae, and not for 3^{rd}–4^{th} instar larvae and for both younger and older *M. pygmaeus* nymphs. Similarly, in no-choice tests on plants, development rate was significantly lower only for *T. absoluta* on plants treated with Mycotal®. In two-choice tests, both pest and predator did not show any preference between fungus-treated and untreated plants. Following the laboratory results, Mycotal® was applied also in plastic tunnel cultivation of tomato under field conditions. The number of *T. absoluta* larvae was significantly lower on plants treated with Mycotal® than on untreated plants, while the presence of *M. pygmaeus* was not affected by the treatment. Therefore, the use of fungal products together with the predator appears to be a promising strategy for the control of *T. absoluta* on tomato crop.

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1. Introduction

The tomato pinworm *Tuta absoluta* (Meyrick) (Lepidoptera: Gelechiidae) [*Phthorimaea absoluta* Meyrick (Chang and Metz, 2021)] is now considered a key tomato pest worldwide (Biondi et al., 2018; Zhang et al., 2020). Native to South America, this pest was first detected in Spain in 2006, and within a few years has become a serious threat to tomato production in most Afro-Eurasian countries, causing heavy economic losses and reaching destructive levels in invaded tomato-growing areas (Bawin et al., 2015; Chailleux et al., 2013; Chouikhi et al., 2022).

After its appearance and spread throughout the Mediterranean basin, chemical control was the only approach used by tomato growers to maintain *T. absoluta* populations below the action threshold (Giorgini et al., 2019), and remains a dominant management tactic, specifically in open-field tomatoes. However, chemical control is made difficult by the cryptic behaviour of the larvae hidden inside mines and the high development and reproduction potential of the species (Desneux et al., 2022). Moreover, the intense use of synthetic chemical insecticides to control this pest has caused the rise of populations resistant to many molecules (Biondi et al., 2018; Giorgini et al., 2019; Konan et al., 2023). Therefore, more sustainable alternative control methods are needed, such as the enhancement and exploitation of natural enemies, which can contribute to the simultaneous control of multiple pests (Konan et al., 2023).

To this end, studies were promptly initiated in the Mediterranean area to identify possible natural enemies capable of adapting to the exotic pest (Ferracini et al., 2019). Larval parasitoids have been identified, such as *Necremnus tutae* Ribes & Bernardo (Hymenoptera: Eulophidae), which is among the most abundant parasitoids found to attack *T. absoluta*, and whose activity to date is promoted through conservation biocontrol methods (Gonthier et al., 2024). In addition, the neotropical parasitoid *Dolichogenidea gelechiidivoris* Marsh (Hymenoptera: Braconidae) has accidentally established following its host in Spain and Algeria, and has been released in a classical biological control programme in Sub-Saharan Africa (Gonthier et al., 2024; Syropoulou et al., 2025). However, attention has shifted to generalist predators that are known to contribute greatly to biological control of many agricultural pests in the Mediterranean area.

In particular, zoophytophagous species of tribe Dicyphini (Hemiptera: Miridae) belonging to the genera *Dicyphus*, *Macrolophus* and *Nesidiocoris* are among the most abundant and effective predators of tomato pests, and have proven suitable candidates as biological control agents in Mediterranean regions (Ingegno et al., 2021). Their effectiveness can be attributed to their voracious consumption of several pests, including eggs and young larvae of *T. absoluta* (Ferracini et al., 2019; Konan et al., 2023), and their zoophytophagous behaviour, which allows them to remain on plants even when prey is scarce (Siscaro et al., 2019; Sylla et al., 2016). Such behaviour is naturally beneficial if with their feeding activity on the plant dicyphine predators do not cause damage, and among these species, *Macrolophus pygmaeus* (Rambur) is considered less harmful and is widely used as a biological control agent (Siscaro et al., 2019; Meesters et al., 2024). In north-western Italy, *M. pygmaeus* is widespread, and is both found naturally colonising tomato crop and used in augmentative releases (Ingegno et al., 2009). Nevertheless, as this species requires alternative prey to support its population growth (Biondi et al., 2018; Giorgini et al., 2019; Sylla et al., 2016), the release of *M. pygmaeus* alone is not always effective in managing *T. absoluta* populations, thus combining it with other biocontrol agents has the potential to improve biological control programmes (Konan et al., 2023).

Among biological control agents, entomopathogenic fungi have proven to be very effective in controlling a wide range of pests (Islam et al., 2021). Some entomopathogenic fungi, among which various isolates of *Beauveria bassiana* (Bals.-Criv.) Vuill., *Metarhizium anisopliae* (Metschnikoff) Sorokin, and *Akanthomyces muscarius* (Petch) Spatafora, Kepler & B.Shrestha [= *Lecanicillium muscarium* (Petch) Zare & W.Gams] (Hypocreales: Cordycipitaceae), have shown to be effective against eggs and 2nd-instar larvae, and have reduced the emergence rate of adults of *T. absoluta*, under laboratory conditions (Abdel-Baky et al., 2021; Aynalem et al., 2022; Askary and Yarmand, 2007; Chouikhi et al., 2022; Cuthbertson et al., 2010; Cuthbertson and Walters, 2005; Klieber and Reineke, 2016; Mohamed Mahmoud et al., 2021). Moreover, recent research has revealed that *B. bassiana* and *M. anisopliae* can be effective against *T. absoluta* not only when present as epiphytes on the surface of tomato leaves, but also as endophytes inside leaves, although with variable results (Agbessenou et al., 2020; Allegrucci et al., 2017; Geremew et al., 2024; Giannoulakis et al., 2023; Klieber and Reineke, 2016; Silva et al., 2020; Zheng et al., 2023). Indeed, entomopathogenic fungi can naturally colonise plant tissues, but they can also be artificially inoculated in various crops (McKinnon et al., 2017; Yerukala et al., 2022; Mantzoukas and Eliopoulos, 2020; Muola et al., 2024; Nishi et al., 2021a). Plant colonisation by entopathogenic fungi can be in some case also beneficial to the crop improving plant growth, enhancing antioxidant capacity and influencing tissue macronutrient and micronutrient contents, or promoting drought tolerance, beyond that reducing pest infestation (Shaalan et al., 2021; Kuzhuppillymyal-Prabhakarankutty et al., 2020; Macuphe et al., 2021). This discovery has broadened the horizons for the interactions between entomopathogenic fungi and plants, and the use of entomopathogenic fungi, especially those with endophytic activity, can be considered a valid alternative to substitute chemical insecticides in sustainable and organic farming system (Mantzoukas and Eliopoulos, 2020; Mohamed Mahmoud et al., 2021; Vidal and Jaber, 2015; Wei et al., 2020 Zheng et al., 2023).

Given their efficacy against various crop pests, *B. bassiana*, *M. anisopliae*, and *A. muscarius* are available as commercial biopesticides and are widely used to control greenhouse pests (Cuthbertson and Walters, 2005; Cuthbertson et al., 2010; Mantzoukas and Eliopoulos, 2020; Prasad and Pal, 2014). Overall, the efficacy of entomopathogenic fungi against *T. absoluta* has been extensively studied, while so far little is known about the effect of these fungi on dicyphine predators. Treatments with *B. bassiana* or *A. muscarium* slightly affected survival of nymphs of the predator *Nesidiocoris tenuis* (Reuter), another commercially available dicyphine species used in biological control, when fungal treatments were applied directly on the nymphs but not when they were applied on tomato plants prior the nymph arrival (Assadi et al., 2021). Concerning *M. pygmaeus*, the few results on both direct and residual effects of the two entomopathogenic fungi are variable. Depending on the commercial product and fungal strain, treatments either did not affect 2nd instar nymph survival in laboratory trials or predator populations in tomato greenhouses (Hamdi et al., 2011), or significantly affected both survival and prey consumption of 5th instar nymphs (Betsi and Perdikis, 2023).

Since *M. pygmaeus* and entomopathogenic fungi are widely used on tomato to control various pests, it appeared crucial to further investigate the impact of commercial formulations of these fungi on the predator *M. pygmaeus*. In particular, the study aimed at evaluating the effects of two commercial formulations, Naturalis® and Mycotal®, based on *B. bassiana* and *A. muscarius*, respectively, authorised for pest control on tomato in Italy. Therefore, the effects of the two products on some biological and behavioural traits of both *T. absoluta* and *M. pygmaeus* were evaluated under both laboratory and field conditions. In the laboratory, the insects were exposed to plants treated with foliar spray, or both foliar spray and soil drenching, in order to investigate whether the two commercial formulations could affect survival, development, and behaviour of the two insects. In the field, populations of *T. absoluta* and *M. pygmaeus* were monitored on plants treated with the laboratory-selected product or untreated to assess the partly direct and partly residual effect on the presence and abundance of both insects. Ultimately, the aim of the study was to evaluate whether fungal application along with predator release on the tomato crop could be a recommendable control strategy to achieve more effective pest control while minimising side effects.

2. Materials and methods

2.1. Insect rearing

Mass rearing of *T. absoluta* started from individuals collected on tomato crop in Piedmont (NW Italy) in summer 2020 and 2021, and maintained continuously in laboratory rearing. *Tuta absoluta* rearing was conducted on tomato plants (*Solanum lycopersicum* L.) in net cages (245 × 245 × 630 mm; BugDorm-4S2260, MegaView Science, Taichung, Taiwan). To obtain groups of larvae of similar age (i.e., 1st-2nd instar larvae and 3rd-4th instar larvae), tomato plants were weekly exposed to *T. absoluta* adults: two new plants were inserted into the cage containing only adults, and after a week of exposure, they were moved to another cage to start the new cohort.

Mass rearing of *M. pygmaeus* started from adults provided by CBC Bioplanet Soc. agr. srl (Cesena, Italy). *Macrolophus pygmaeus* rearing was conducted on tobacco plants (*Nicotiana tabacum* L.) in net cages (475 × 475 × 475 mm; BugDorm-4S4590, MegaView Science, Taichung, Taiwan), supplied with eggs of *Ephestia kuehniella* Zeller (Lepidoptera: Piralidae) and cysts of *Artemia salina* (L.) (Anostraca: Artemiidae) as alternative prey. To obtain groups of nymphs of similar age (i.e., 1st-2nd instar nymphs and 3rd-4th instar nymphs), flat bean pods (*Phaseolus vulgaris* L.) were exposed twice a week to *M. pygmaeus* adults: two–three pods were inserted into the cage, and after three days of exposure, they were moved to a 3 l-plastic box closed with a removable lid (265 × 175 mm), and with holes on the sides closed with insect-proof net, to start the new cohort.

Mass rearing of both pest and predator was maintained in climatic chambers at 24 ± 1 °C, 65 ± 5 % RH, and 16L:8D.

2.2. Plant cultivation and treatment

For the trials, approximately 20 cm high tomato plants *S. lycopersicum* cv 'Cuore di bue' were used. Plants were cultivated in an experimental greenhouse at 27 ± 3 °C and 55 ± 23 % RH. Seeds were sown in plastic pots (Ø 30 mm) with vermiculite and soil, and watered daily; when seedlings had reached 4 cm in height, they were transferred to bigger plastic pots (Ø 70 mm).

Before using them in the trials, tomato plants were treated with: i) Naturalis® [*B. bassiana* ATCC 74040, oil based suspension concentrate containing 2.3×10^7 CFU (viable spores) ml^{-1}, Biogard, CBC (Europe) S. r.L., Grassobbio, Italy], at label dose 1 ml l^{-1}, ii) Mycotal® (*A. muscarius* Ve6 water dispersible granules containing 1×10^{10} CFU (viable spores) g^{-1}, Koppert Italia S.r.L., Bussolengo, Italy], at label dose 1 g l^{-1}, and iii) water as untreated control. Depending on whether the trial was conducted on a leaflet or on a whole plant, plants were treated with the fungal product by foliar spraying only, or by two soil applications 4 days apart, followed by a foliar spray 3 days after the second soil application.

2.3. No-choice tests on leaflets

Trials to evaluate mortality of *T. absoluta* larvae and *M. pygmaeus* nymphs were performed in Petri dishes (Ø 15 cm). Apical tomato leaflets were cut from plants, and petioles were inserted into a 2 ml plastic tube filled with water and sealed with Parafilm® to maintain leaf turgor during the experiments. Then, they were sprayed with 1 ml of fungal suspension or water by a glass hand sprayer, and individually placed into a Petri dish. When the leaflets were dried, five individuals of the same cohort of *T. absoluta* or of *M. pygmaeus* were gently transferred, using a thin brush under a stereomicroscope, onto the leaflets subjected to the following treatments in comparison: i) treated with Naturalis®, ii) treated with Mycotal®, and iii) treated with water (control). Trials was carried out with: i) *T. absoluta* 1st-2nd instar larvae, ii) *T. absoluta* 3rd-4th instar larvae), iii) *M. pygmaeus* 1st-2nd instar nymphs, and iv) *M. pygmaeus* 3rd-4th instar nymphs. For each combination of treatment, species and stage, five replications were performed, and the experiment was repeated five times (for a total of 25 replications per each combination of treatment, species and stage). Mortality rate was assessed after 7 days. All trials were performed in a climatic chamber at 24 ± 1 °C, 65 ± 10 % RH and 16:8 L:D.

2.4. No-choice tests on plants

Trials to evaluate development rate of *T. absoluta* and *M. pygmaeus* were carried out on whole plants treated with i) Naturalis®, ii) Mycotal®, iii) water (control), as described in 2.2. Insects were placed on plants 24 h after the foliar spray.

For *T. absoluta*, five eggs per plant were gently transferred, using a thin brush under a stereomicroscope, from leaves where they were laid in the laboratory rearing to the treated plants. Each plant was then inserted into a net cage (245 × 245 × 630 mm; BugDorm-4S2260, MegaView Science, Taichung, Taiwan). For each treatment, 10 plants were used, and the experiment was repeated five times (for a total of 50 replications per treatment). All cages containing treated plants with eggs were placed separately for each treatment in structures (one per treatment) supported by a stainless-steel frame (1.5 × 1.5 × 1.1 m) and closed with an insect-proof net, in the experimental greenhouse at 27 ± 3 °C and 55 ± 23 % RH. Each plant was checked every 2 days to record adult emergence.

For *M. pygmaeus*, treated plants were isolated individually inside glass cylinders (Ø 80 mm, height 150 mm), closed at the top with net, and supplied with eggs of *E. kuehniella*. Five 1st instar nymphs per plant were transferred from the laboratory rearing to the plant inside each cylinder. For each treatment, 10 plants were used, and the experiment was repeated four times (for a total of 40 replications per treatment). All isolated plants with nymphs were then placed separately for each treatment in structures (one per treatment), supported by a stainless-steel frame (1.5 × 1.5 × 1.1 m) and closed with an insect-proof net (mesh 0.23 × 0.23), in the experimental greenhouse at 27 ± 3 °C, 55 ± 23 % RH, and 12L:12D. Each plant was checked every 2 days to record adult emergence.

2.5. Two-choice tests with plants

To assess the behavioural responses of *T. absoluta* and *M. pygmaeus* to fungus-treated or untreated tomato plants, two-choice tests were performed using whole plants treated as described in 2.2. Plants were used 24 h after the foliar spray. The following three combinations were tested: i) plant treated with Naturalis® *vs* plant treated with water (control), ii) plant treated with Mycotal® *vs* plant treated with water (control), and iii) plant treated with Naturalis® *vs* plant treated with Mycotal®.

For *T. absoluta*, oviposition preference was assessed by exposing two differently treated tomato plants to 10 adults for 72 h. The plants were placed, so as not to be in contact, inside a cage resting on its longest side (630 × 245 × 245 mm; BugDorm-4S2260, MegaView Science, Taichung, Taiwan), in the experimental greenhouse at 27 ± 3 °C and 55 ± 23 % RH. After 72 h, plants were transferred to the laboratory, and eggs laid on each plant were counted under a stereomicroscope. For each combination, 25 replicates were carried out.

For *M. pygmaeus*, behavioural response of females to the odours of differently treated plants was assessed by olfactometer bioassays. Before using them, *M. pygmaeus* females were individually kept without any prey or plant in a glass tube (length 120 mm, Ø 23 mm) closed with a humid cotton cap for 24 h. The bioassays were carried out in a vertical Y-shaped Pyrex tube (internal Ø 23 mm) formed by an entry arm 250 mm long, and two side arms 200 mm long (70° angle between arms), and positioned vertically, and the setup of the trials was the same as in Leman et al. (2020). Each female was observed until it had walked at least 60 mm up one of the side arms or until 10 min had elapsed; if it did not choose a side arm within 10 min, the female was considered as "no choice" and not included in the subsequent data analysis. Each female was evaluated only once to prevent behaviour from being affected by experience. After testing 10 responses, the volatile sources were replaced with a new set of plants, so four sets of plants were tested per pair of volatile sources (4 replicates per experiment, 40 females per combination). The olfactometer bioassays were conducted at 25 ± 2 °C, 55 ± 2 % RH, and 350 lx.

2.6. Field trial

Following the results obtained in the laboratory, the effect of Mycotal® use was also evaluated in the field on the tomato crop at a farm located in Moretta (Piedmont, NW Italy; 44.7823 N, 7.5155E, 250 m a.s.L.) in 2021. Tomato plants of cv 'Cuore di Bue' (hybrid Meneghino®, Rijk Zwaan Italia srl, Bologna, Italy) were transplanted in mulched soil in May; throughout the season, they were irrigated with a drip irrigation system and managed according to standard local cultural practices. The trial was conducted on tomato plants grown in a 540 m^2 plastic tunnel (60 × 8 m).

The tunnel was divided in eight plots, and four plots were randomly assigned to each of the two treatments, which were: i) application of Mycotal® at the label dose, and ii) no insecticide application as control. Mycotal® was applied twice by soil application in tray (400 ml $tray^{-1}$), two weeks before transplanting (April 28) and at the time of transplanting (May 13), and after one month, three times by foliar spray (100 ml m^{-2}) every two weeks (June 9 to July 8). Starting from mid-June, every 10 days the presence of *T. absoluta* larvae and of *M. pygmaeus* mobile stages (both juveniles and adults) was checked on tomato plants until mid-September, for a total of nine surveys. For *T. absoluta*, in each survey and in each plot, three different plants were randomly selected (i.e., 12 plants per treatment), and for each plant, five randomly selected leaves were visually inspected to count active larvae inside the mines. A total of 60 leaves per treatment were inspected in each survey. For *M. pygmaeus*, in each survey and in each plot, five different plants were randomly selected (i.e., 20 plants per treatment), and for each plant, leaves were beaten on a tray (240 × 340 mm) at three levels of the plant: apical, median, basal. All individuals fallen on the tray at each plant level were identified and counted. A total of 60 tray beats per treatment were made in each survey.

2.7. Statistical analysis

In no-choice tests, proportions of dead individuals per leaflet inside Petri dish and of emerged adults per plant in cage or cylinder were compared, separately per species and stage, with a binomial distribution with the generalized linear model (GLM) using a logit link function, with treatment, replicate and their interaction as factors. In the case of significant differences, means were then separated at $P < 0.05$ using the Bonferroni test under the GLM procedure.

In two-choice trials, the mean proportion of eggs laid on the two plants inside each cage, and the proportion of insects responding to one of the two volatile sources were compared with a 50 % distribution. The significance of the response was tested using a Chi-square (χ^2) test.

In the field trial, the total number per plot in each survey of *T. absoluta* larvae and of *M. pygmaeus* mobile stages was compared with a Poisson distribution with the GLM using a log link function, with treatment, date and their interaction as factors. Surveys in which no *T. absoluta* larvae or no *M. pygmaeus* individuals were found, even in the four plots of a single treatment, were discarded from the comparison.

All statistical analyses were performed using the software IBM SPSS® Statistics 28.0.1 (IBM Corp., NY, USA).

3. Results

3.1. No-choice tests on leaflets

By statistical analysis, no interaction between treatments and replicates in any species or stage was found, namely: i) 1st-2nd instar *T. absoluta* larvae, GLM, $\chi^2 = 12.758$, df = 8, P = 0.120; ii) 3rd-4th instar *T. absoluta* larvae, GLM, $\chi^2 = 13.196$, df = 8, P = 0.105; iii) 1st-2nd instar *M. pygmaeus* nymphs, GLM, $\chi^2 = 7.723$, df = 7, P = 0.358; iv) 3rd-4th instar *M. pygmaeus* nymphs, GLM, $\chi^2 = 6.163$, df = 8, P = 0.629. Instead, treatment had a significant effect on mortality of 1st-2nd instar *T. absoluta* larvae, which was significantly higher on individuals feeding on leaflets treated with Naturalis® or with Mycotal® than those feeding on untreated leaflets (Table 1). For 3rd-4th instar *T. absoluta* larvae as well as for *M. pygmaeus* nymphs of different instar, treatment had no significant effect on mortality (Table 1).

3.2. No-choice tests on plants

No interaction between treatments and replicates in any species was found, namely: i) *T. absoluta*, GLM, $\chi^2 = 11.487$, df = 8, P = 0.176; ii) *M. pygmaeus*, GLM, $\chi^2 = 10.261$, df = 6, P = 0.114. Instead, treatment had a significant effect on *T. absoluta* adult emergence, which was significantly lower for individuals feeding on tomato plants treated with Mycotal® than those feeding on tomato plants treated with Naturalis® and with water (Table 2). For *M. pygmaeus*, treatment did not show any significant effect on adult emergence rate (Table 2).

3.3. Two-choice tests with plants

Replicates in which the total number of laid eggs was less than 10 were not considered in the statistical analysis, i.e., from 2 to 5 cages per combination. Overall, 3,124 *T. absoluta* eggs were laid in the cages included in the statistical analysis, with an average of 53.9 ± 4.9 per cage. No significant preference was observed in *T. absoluta* oviposition in any comparison. When plants treated with Mycotal® were compared to control plants, *T. absoluta* laid on average 57.3 %±5.3 of eggs on plants treated with fungal product ($\chi^2 = 2.137$; df = 1; P = 0.144; total number of eggs 749). When plants treated with Naturalis® were compared to control plants, *T. absoluta* laid on average 42.7 %±5.0 eggs on plants treated with fungal product ($\chi^2 = 2.137$; df = 1; P = 0.144; total number of eggs 1,184). In the comparison between the two fungal products results were similar, with 49.1 % ± 6.4 of eggs laid on plants treated with Mycotal® and 50.9 %±6.4 of eggs laid on plants treated with Naturalis® ($\chi^2 = 0.029$; df = 1; P = 0.865; total number of eggs 1,269).

Similarly, *M. pygmaeus* showed no significant preference in any comparison. When plants treated with Mycotal® were compared to control plants, 51.7 % of *M. pygmaeus* females chose the control plant

Table 1
Mean mortality (percentage ± SE) of 1st-2nd instar and 3rd-4th instar larvae of *Tuta absoluta*, and 1st-2nd instar and 3rd-4th instar nymphs of *Macrolophus pygmaeus*, after one week of exposure to tomato leaflets treated with fungal suspension or water (control). Means within a column followed by the same letter are not significantly different [Bonferroni test: P < 0.05, under GLM procedure with binomial distribution and logit link].

Treatment	Mortality (%)			
	Tuta absoluta		*Macrolophus pygmaeus*	
	L 1–2	L 3–4	N 1–2	N 3–4
Naturalis®	40.0 ± 4.9 a	31.2 ± 4.0 a	36.0 ± 6.1 a	31.2 ± 3.8 a
Mycotal®	43.2 ± 6.1 a	21.6 ± 3.4 a	28.8 ± 5.0 a	23.2 ± 4.1 a
Control	25.6 ± 3.4 b	28.8 ± 4.3 a	34.4 ± 5.7 a	20.8 ± 3.6 a
GLM				
χ^2	9.784	2.246	1.609	3.910
df	2	2	2	2
P	0.008	0.325	0.447	0.142

Table 2
Mean proportion (±SE) of emerged adults of *Tuta absoluta* and *Macrolophus pygmaeus* on tomato plants treated with fungal suspension or water (control). Means within a column followed by the same letter are not significantly different [Bonferroni test: P < 0.05, under GLM procedure with binomial distribution and logit link].

Treatment	Adult emergence rate (%)	
	Tuta absoluta	*Macrolophus pygmaeus*
Naturalis®	36.4 ± 2.90 a	66.5 ± 4.44 a
Mycotal®	26.0 ± 2.87 b	63.0 ± 4.94 a
Control	36.0 ± 3.13 a	72.0 ± 4.57 a
GLM		
χ^2	8.046	3.488
df	2	2
P	0.018	0.175

(χ^2 = 0.118; df = 1; P = 0.731; 38 responding females); when plants treated with Naturalis® were compared to control plants, 54.3 % of *M. pygmaeus* females chose the control plant (χ^2 = 0.735; df = 1; P = 0.391; 35 responding females); when plants treated with Mycotal® were compared with plants treated with Naturalis®, 53.8 % of *M. pygmaeus* females chose the plant treated with Mycotal® (χ^2 = 0.593; df = 1; P = 0.441; 39 responding females). The number of non-responding females was generally low, from 1 to 5 per combination.

3.4. Field trial

The mean number of *T. absoluta* larvae and of *M. pygmaeus* mobile stages sampled in each plot in each survey is shown in Figs. 1–2. *Tuta absoluta* larvae were first found on July 5, in small numbers, and only in the control; then, starting from July 26 (two weeks after the last foliar spray with Micotal®), they were found in all surveys and in both treatments, with an increasing trend. *Macrolophus pygmaeus* individuals were already observed in the first survey, with small numbers increasing in the last two surveys. Overall, the mean number of *T. absoluta* larvae was significantly lower in the plots treated with Mycotal® than in the control plots (Table 3), with no interaction between date and treatment (GLM, χ^2 = 2.224, df = 4, P = 0.695), but with differences in abundance in relation to date (GLM, χ^2 = 42.439, df = 5, P < 0.001) (Fig. 1). For *M. pygmaeus*, conversely, the treatment had no effect on its occurrence (Table 3), with no interaction between date and treatment (GLM, χ^2 = 2.793, df = 5, P = 0.732), but with differences in abundance in relation to date (GLM, χ^2 = 21.669, df = 5, P < 0.001), as the presence was not constant during the season (Fig. 2).

4. Discussion

The effect of entomopathogenic fungi on *T. absoluta* has been investigated in various studies conducted, however, with different

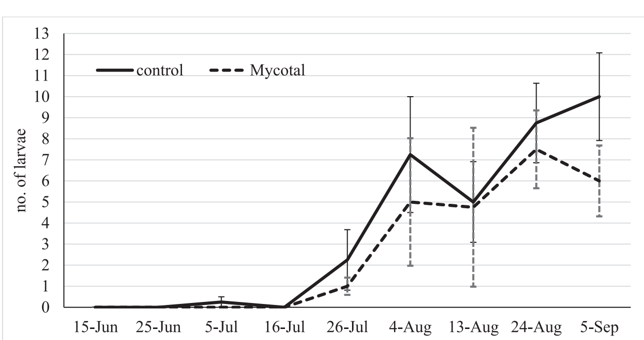

Fig. 1. Mean number (±SE) of larvae of *Tuta absoluta*, counted per plot in each survey in the field trial in the plots treated with Mycotal® or untreated (control).

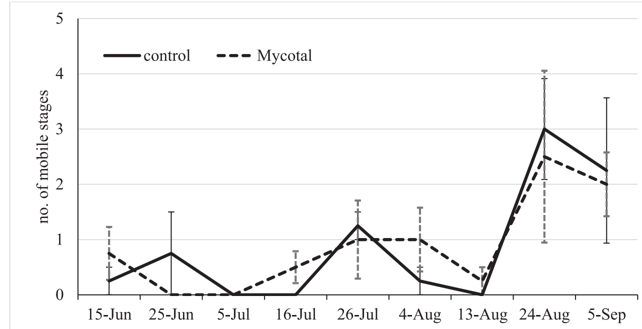

Fig. 2. Mean number (±SE) of mobile stages of *Macrolophus pygmaeus*, counted per plot in each survey in the field trial in the plots treated with Mycotal® or untreated (control).

Table 3
Mean number (±SE) of larvae of *Tuta absoluta* and of specimens of *Macrolophus pygmaeus* per plot and per survey in the field trial, in the plots treated with Mycotal® or untreated (control). Means within a column followed by the same letter are not significantly different [Bonferroni test: P < 0.05, under GLM procedure with Poisson distribution and log link].

Treatment	*Tuta absoluta* (no. of larvae)	*Macrolophus pygmaeus* (no. of mobile stages)
Mycotal®	4.00 ± 0.98 a	1.33 ± 0.32 a
Control	5.58 ± 0.99 b	1.29 ± 0.34 a
GLM		
χ^2	5.510	0.934
df	1	1
P	0.019	0.334

strains, doses, and modes of application, and thus leading to sometimes conflicting results. Here, we aimed to evaluate the activity of two commercial fungal products on biological and behavioural aspects of the pest and the predator widely used for its control, in order to provide useful data for implementing a pest management strategy based on the joint use of biological control agents.

In no-choice tests on leaflets with Naturalis®, our results are consistent with those of other studies focusing on the efficacy of *B. bassiana* against *T. absoluta*, confirming the vulnerability of younger larvae (Allegrucci et al., 2017; Chouikhi et al., 2022; Silva et al., 2020). In the other studies, however, mortality of *T. absoluta* larvae was usually higher, but it should be considered that different concentrations of conidia or spores in the solution, different fungal strains, and also different application methods were used. Indeed, mortality of 2nd instar *T. absoluta* larvae reached values of 87.5 ± 5 % when placed on leaves previously immersed in a fungal suspension with 10^8 conidia ml^{-1} of *B. bassiana* strain LPSC 1067 (Allegrucci et al., 2017), and mortality rates of 3rd and 4th instar *T. absoluta* larvae were 100 % and 55 % with the highest and lowest fungal concentration, respectively, but in this case the suspension of *B. bassiana* strain Sn182 (10^5, 10^6 and 10^7 spores ml^{-1}) was applied directly on larvae (Mohamed Mahmoud et al., 2021). In our study, the fungal suspension of the commercial strain at the label dose (10^7 spores ml^{-1}) was applied by spraying on the leaflets, and *T. absoluta* larvae were placed on leaflets only after the treatment. Using the same commercial formulation of *B. bassiana* ATCC 74040 (Naturalis ®) and the same mode of exposure, mortality of 2nd instar larvae was similar to that obtained in our study, and increased only increasing the concentration of the fungal suspension (Chouikhi et al., 2022). Similarly, for the commercial formulation of *A. muscarius* Ve6 (Mycotal®), our results were consistent with those obtained using the same fungal strain and mode of exposure of 2nd instar *T. absoluta* larvae (Chouikhi et al., 2022), which was confirmed to be the most susceptible stage to exposure of contaminated substrate. Higher mortality rates than ours were obtained only by increasing the concentration of the fungal suspension or by directly treating the leaflets bearing active mines, i.e., within which 2nd instar *T. absoluta* larvae were present (Chouikhi et al., 2022).

In contrast, the survival of both 1st-2nd instar and 3rd-4th instar *M. pygmaeus* nymphs was not affected by the fungal treatment applied on leaflets, consistent to what was observed for 2nd instar nymphs after direct exposure or exposure to contaminated substrate and prey, *Trialeurodes vaporariorum* (Westwood) (Hemiptera: Aleyrodidae), with both fungi (Hamdi et al., 2011). In another study, however, *B. bassiana* was slightly harmful to 5th instar *M. pygmaeus* nymphs after both direct exposure and contact with contaminated substrate, but this could be due to a variation in the sensitivity of older nymphal instars to *B. bassiana*, or to higher conidial concentration applied (Betsi and Perdikis, 2023). Younger *M. pygmaeus* nymphs (1st to 4th instar) seem therefore to be less susceptible to fungal contamination than older nymphs (5th instar) and *N. tenuis* nymphs. Mortality of *N. tenuis* nymphs was found to be slightly increased (about 10 %) both when nymphs were treated directly with the fungal suspensions, and when nymphs were introduced on plants infested by *Bemisia tabaci* (Gennadius) (Hemiptera: Aleyrodidae) already treated with fungal suspensions, one based on *A. muscarius* strain Ve6 (19–79) with a water-soluble granular formulation of 10^{10} spores g^{-1} (as for Mycotal®), and the other based on *B. bassiana* strain R444 wettable powder with a concentration of 2×10^9 conidia g^{-1} (different from Naturalis®) (Assadi et al., 2021).

In no-choice tests on plants, only on plants treated with Mycotal® the development rate of *T. absoluta* from egg to adult was reduced, by about one third, while that of *M. pygmaeus* was not affected by any treatment. On the contrary, tomato plant inoculation with *B. bassiana* Crete isolate, *M. anisopliae* Crete isolate, or the commercial *B. bassiana* strain GHA 10.735 %, negatively affected *T. absoluta* development (Giannoulakis et al., 2023). This difference could be due to the different strains of *B. bassiana*, or the inoculation protocol.

Further promising results emerged from two-choice tests on plants; in fact, both fungal products did not appear to affect the behavioural responses of both *T. absoluta* females, which laid eggs equally on treated and untreated plants, and *M. pygmaeus* females, which indifferently chose fungus-treated or untreated plants in olfactometry bioassays. Similarly, in a two-choice cage assay *M. pygmaeus* was shown to choose indifferently tomato plants inoculated or not inoculated with *B. bassiana* ARSEF 3097 or *Metarhizium brunneum* Petch strain ARSEF 1095 (Meesters et al., 2024). Furthermore, in an olfactometry bioassay, *Macrolophus basicornis* (Stal) (Hemiptera: Miridae) choose indifferently tomato plants untreated or treated with *Metarhizium robertsii* J.F.Bisch., S.A.Rehner & Humber strain 'ESALQ 1635′ (Salazar-Mendoza et al., 2024). In the same study, *T. absoluta* showed to prefer to lay eggs on untreated tomato plants compared to plants treated with *M. robertsii* strain 'ESALQ 1635′ (Salazar-Mendoza et al., 2024), contrary to what was observed in our study with the two commercial formulations based on *B. bassiana* and *A. muscarius*. Therefore, *Macrolophus* species, in particular *M. pygmaeus*, seem to be less disturbed from the fungal treatments on tomato plants, as well as *T. absoluta*, when the two selected commercial formulations are used. This could be doubly advantageous, since *T. absoluta* developing on treated plants will have increased mortality of younger larvae, and if Mycotal® is used, also a reduced development rate, and it will also be preyed by undisturbed *M. pygmaeus*, resulting in an additive effect on population reduction.

Since both commercial products were found not to affect *M. pygmaeus* activity and increase mortality of younger larvae of *T. absoluta*, but only Mycotal® was also shown to have an effect on development from eggs to adults, this product was selected for the field trial. Following the label recommendations, three treatments were applied, and in addition two soil applications before transplanting and at transplanting were made as a strategy to control pathogens as well (Jaber and Ownley, 2018). Confirming what was observed in the laboratory, in the field presence of mobile stages of *M. pygmaeus* on plants was not affected by the treatments with Mycotal®: the predator population fluctuated without any significant differences between treated and untreated plots, even following direct fungal treatments early in the season. Therefore, as the predator is not affected by the commercial formulation, it might be able to establish on the crop after the treatment and increase its population to contrast the pest for a long time. Thus, the application of Mycotal® on the tomato crop seems to be compatible with the use of the predator, and even additive.

In contrast, in the field trial, the presence of active *T. absoluta* larvae was lower on treated plants. However, pest population was observed on tomato mainly starting from mid-July, i.e., two weeks after the last foliar spray and two months after the last soil application; therefore, the effect of Mycotal® could be due to the *A. muscarius* epiphytic development on the leaf surface, considering also that it is not yet known whether it can also colonise tomato plant tissues as an endophyte. Indeed, it was shown that *A. muscarius* IMI 268317 had epiphytic growth on tomato leaf surface as early as one day after treatment, which continued until the end of the trial (14 days), whereas no endophytic colonisation of tomato tissues was observed for the entire period (Nishi et al., 2021b). However, the relative *Akanthomyces lecanii* (Zimm.) Spatafora, Kepler & B.Shrestha was found to succesfully colonise tomato leaves as early as one day after the foliar spray, and throughout the investigation period (21 days) (Gurulingappa et al. 2010). Therefore, in light of our results, *A. muscarius* was shown to affect *T. absoluta* mortality and development over a long period of time, leading to the hypothesis that it can succesfully colonise tomato leaf surface after foliar application even in the field. However, further studies will be needed to assess how and how long *A. muscarius* can colonise tomato plant tissues.

5. Conclusions

In our study, some effectiveness of the two commercial formulations Naturalis® and Mycotal®, registered for application on tomato crop, was observed against the younger larvae of *T. absoluta* when exposed to treated tomato leaves in laboratory tests. Moreover, these formulations showed no residual effects on the survival of the predator *M. pygmaeus*, as well as its choice in olfactometry bioassays, or on the oviposition rate of *T. absoluta* on treated or untreated plants in two-choice laboratory tests. In addition, Mycotal® had no effects on *M. pygmaeus* presence in the field trial, while it slightly reduced active *T. absoluta* larvae inside mines. Thus, the use of Mycotal® seems to be compatible with the release of *M. pygmaeus* on tomato crop, with an additive effect on reducing *T. absoluta* population. Consequently, further investigation will be needed to implement the most effective strategy in terms of the number and timing of application of treatments.

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CRediT authorship contribution statement

Sara Scovero: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Silvia T. Moraglio:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Luciana Tavella:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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